Beyond Codepoint CER: A Cross-Generation Benchmark for Real-World Tamil OCR

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Abstract— ***Benchmarking of Tamil OCR systems is performed on prepared datasets of pristine images using codepoint-level Character Error Rate (CER), which is simply lifted from Latin-script evaluation process. In this paper, we claim that this paradigm of benchmarking is biased towards understanding of the OCR performance for Tamil, abugida script in which matras and conjuncts lead to mapping of one visual akshara to multiple graphemes and, therefore, codepoint-level metric confuses simple grapheme order change with error. We propose a two-part benchmarking paradigm: Benchmarking & Data part providing preparation of stratified set of real-life Tamil documents annotated across four dimensions (matra presence, conjunct number, degradation type, and capture technique) and evaluation of seven OCR baselines from three generations of technology in the same no-paid-API setup; and Metrics part computing grapheme-cluster-aware CER/WER and reorder/misread-separated F1 score as CORE metrics. Our results from the first run based on IndicSTR12 (n=100) will be presented along with the results of the core benchmarking run, based on all seven baseline systems applied to 644 images chosen from five datasets of Tamil text that are either real world or synthetic, as well as the publicly available leaderboard mechanism that allows to benchmark new OCR engines against the same criteria. The second paper related to this topic [16] uses the created benchmark to explain the reasons behind failure of particular systems; the current paper is intentionally focused on benchmarking itself and CORE metrics in order for both papers to stand on their own.***

Index Terms—Tamil OCR, grapheme-aware evaluation, akshara segmentation, character error rate, benchmark across generations, OCR leaderboards, low resource script evaluation, Indic scripts

I. INTRODUCTION

The digitalization of printed and handwritten material is an important technology that enables access to educational content and government services as well as documents in Tamil Nadu and other regions where Tamil is spoken. Government documents, land registers, educational documents in schools and colleges, library archives, and common signboards are all digitized with the expectation that OCR will enable their search and translation into machine readable form. OCR is not a secondary but an enabling technology, and its practical feasibility in Tamil is an important factor in the issue at hand.

The Tamil script is an abugida in which each consonant carries a default vowel sound that may be modified by matras in the form of vowel marks that are located before or after the consonant in question. The use of viramas in the Tamil script leads to many cases where more than one consonant is represented by a single akshara. These two characteristics mean that the akshara's representation visually does not map one-to-one to a Unicode code point in many cases, nor does it map to a Unicode grapheme cluster. To this is added the noise associated with real-world document image acquisition in the form of rotation or skewing of the document, low contrast of the image due to aged paper or insufficient illumination, blurring from hand-held document acquisition, and the use of non-Unicode fonts in Tamil printed documents.

The current evaluation procedure of Tamil Optical Character Recognition (OCR), however, is based on measuring just one CER for rather clean benchmark images at the level of codepoints. This approach suffers from two significant drawbacks. The first drawback is evident since the current evaluation process does not reflect practical capturing conditions mentioned above, as rather clean benchmark images are not related to scanning of any government and educational documents. The second drawback is not very apparent, but much more critical, since the CER at the level of codepoints cannot differentiate between the case when the recognition produced errors and the case when the result occurred because of different akshara orders in Unicode representation and image of akshara string. These are the four main contributions of this paper in this context. First, we define a stratified benchmark of realistic Tamil texts that are annotated with tags on the basis of four different variations, such as matras, conjunct density, kind of corruption, and channel of acquisition, in order to test the performance of OCR systems separately for each stratum and not in terms of an overall performance across all strata. Second, we define encoding order-aware CORE metrics, namely, CER, WER, and an F1 metric in terms of reorderings and misreads, whose exact formulation is defined in Section III-D. Next, we assess seven OCR systems as baselines using pre-existing models, representing three generations of technologies, on the stratified benchmark at pilot scale (n=100) and corpus scale (n=644 from five data sets). Finally, we present a leaderboard interface wherein a new OCR engine, available either as an HTTP service or a model file, is assessed via the same protocol. The second contribution towards our goal in this regard is the

diagnostic paper [16] which utilizes the same benchmark and metrics interface to determine the cause for the failure of each method through confusion pair taxonomies and cross-system error correlation at the akshara level.

The organization of the rest of this paper is as follows. Section II describes related works. Section III discusses the system design proposed, including the exact CORE metric formulas and leaderboard interface. Section IV describes the pipeline implemented. Section V gives results from the pilot and benchmark leaderboard. Section VI deals with future work, while Section VII is the conclusion.

II. LITERATURE SURVEY

Previous studies in Tamil OCR have proved the feasibility of automating Tamil characters recognition. The Tamil Optical Character Recognition system has been developed in [11]. An industrial quality Tamil OCR capable of handling independent font and size text has been realized in [12]. In both instances, it has been demonstrated that the automated recognition of Tamil characters is possible using classical feature-based methods as well as the neural network model, but all of these were done on small and relatively noise-free collections of images, whereas in the current study we consider more complex and varied degraded real-world captures.

The dataset uTHCD, introduced in [2], has provided a significant increase in the quantity of Tamil handwritten character data useful for Tamil handwriting recognition because of the lack of such data. Even though uTHCD dataset has provided a considerable increase in the quantity of Tamil handwriting annotated data, it has been done for isolated characters and words, not full documents, and does not discriminate samples based on degradation types such as skew, blur, and contrast loss.

Regarding scene-text benchmark level, the IndicSTR12 [4] benchmark created by CVIT, IIIT Hyderabad, has collected more than 27,000 images of real word and more than 3 million synthetic images per language for twelve major Indian languages including Tamil, and tested the state-of-the-art models like PARSeq, CRNN, and STARNet on the IndicSTR12 benchmark. The above facts clearly indicate the high popularity of realistic wild scene text benchmark data, but IndicSTR12 involves isolated word-level scene text data without degradation and script structural stratification of the benchmark discussed here. Another data set in this domain, IndicDLP [3], is a large-scale multi-lingual and multi-domain document layout parsing data set.

A natural choice of baseline candidates are the existing general-purpose multilingual OCR and vision-language models. While PaddleOCR-VL [6] is a small 0.9 billion parameter model demonstrating strong multilingual document parsing ability, E-ARMOR [7] tests multilingual OCR models in realistic, potentially adversarial scenarios as opposed to carefully crafted benchmarks. When it comes to large multimodal models, OCRBench [13] and its follow-up, OCRBench v2 [14], provide a perfect example of how OCR capabilities should be benchmarked, however they remain highly biased toward Latin and Chinese script based content. The most relevant recent works could be found in GlotOCR Bench [5], where it is shown that "current models still struggle beyond a handful of Unicode scripts" with Tamil being one of the scripts where model's performance degrades, thereby validating the motivation of this paper, even though the evaluation is done on codepoints.

Speaking about metrics, the nearest one in the comparison with the presented CORE metric layer is grapheme-kit [1], to the best of our knowledge: grapheme-kit extends CER/WER-related metrics by taking into account graphemes in Unicode rather than characters and explicitly includes Tamil and Sinhala. Grapheme-cluster splitting is performed in our approach as in [1], which is enhanced with reorder/misread decomposition of the F1-score in Section III-D. As we have seen from the literature review, there is indeed a significant gap that the paper attempts to cover: existing datasets related to Tamil/Indic languages deal with recognition of characters/words [2],[4] or page layout [3], but do not provide datasets where degradation of documents with a stratified approach is considered, especially in view of the recent multilingual datasets [5],[13],[14].

III. PROPOSED MODEL

In this paper, we propose a dual component approach, where Benchmarking & Data Component is employed for building the stratified and realistic Tamil document dataset and processing it with seven best available OCR systems over three technological generations in the exact same manner, and Metric Component assesses their performance in terms of grapheme-sensitive CORE metrics. The sixth step introduces a public leaderboard which allows new metric engines to be benchmarked in the exact same manner.

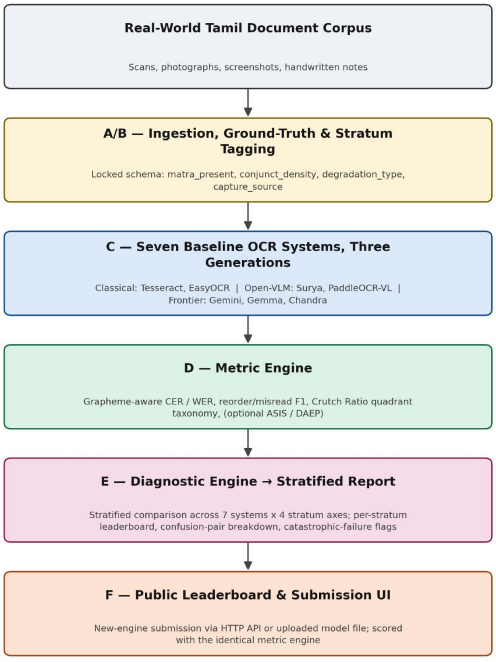


Fig. 1. Proposed benchmarking and public-leaderboard pipeline. Letters correspond to subsections A–F. (A structural-diagnosis extension of stage D/E is developed in a companion paper [16].)

A. DATASET

This corpus is made up of authentic documents in the Tamil language and not artificially constructed: government circulars and notices scanned, textbook and library pages printed, photo taken of signs and scene text, and handwriting sampled using the camera. The annotation of each text document in the corpus is done using a consistent four-strata stratification process determined during the design phase of this project, which include matra_present (whether there are dependent vowels in the akshara level text), conjunct_density (low, medium, and high density according to virama connected consonants), degradation_type (clean, skew, low contrast, blur, and occlusion), and capture_source (scan, camera, and screenshot). There is a minimum number of samples per stratum, and strata that do not meet this minimum will be tagged accordingly and merged with other strata.

B. PRE-PROCESSING

Normalization of format precedes double-keyed transcription of the documents with the resolving of any inconsistencies that might arise and strata tagging with automatic tags verified and sealed by hand. In our case, the licensing requirement is a hard gate and only those documents whose license clearly specifies the rights to reuse enter the corpus.

C. BASELINE OCR SYSTEMS

The list of seven OCR models, which have been created in three different technological generations and are all evaluated in the same way based on the stratified corpus, provided that no commercial API and no tuning for any particular model exist. In the first generation of classical and open-source models, there are Tesseract OCR with the Tamil trained data model and the EasyOCR with its Tamil recognition model. In the second generation of open vision-language models, there are Surya-class multilingual OCR system and PaddleOCR-VL [6].

D. METRIC ENGINE

There are two types of CORE measures that can be calculated through the Metric Engine. These metrics are grapheme cluster-based, and therefore are unaffected by the increased number of errors because of pre-base matra reordering as they are Unicode grapheme cluster-based and not code point-based. The definition of the Grapheme-aware Character Error Rate is:

$$\mathrm{CER} = \frac{\mathrm{ED}(H_g, R_g)}{N_g} \tag{1}$$

where ED(·, ·) is the Levenshtein edit distance, Hw/Rw are the hypothesis/reference word sequences, and Nw = |Rw|. The Word Error Rate can be computed exactly similarly over word sequences split on whitespaces:

$$\mathrm{WER} = \frac{\mathrm{ED}(H_w, R_w)}{N_w} \tag{2}$$

F1 score is composed of two types of scores, namely the reorder score and the misread score. It helps differentiate the errors in encoding orders from genuine mistakes:

$$F1_{\text{misread}} = \frac{M}{N}, \; F1_{\text{reorder}} = \frac{R}{N} \tag{3}$$

where N is the reference length in graphemes, R is the number of edit operations which can be classified into one of the recognized pre-base matra/consonant reordering cases, and M is the number of all other edit operations. The decomposition is accurate because the sum of R and M is exactly equal to the number of edits, meaning that by definition $F1_{\text{reorder}} + F1_{\text{misread}}$ = CER. Another paper [16] which is complementing this one, presents operationalization of exactly this kind of decomposition in terms of five-category classification of confusion pairs, and extends its description by including optional structural integrity metrics for aksharas. This aspect is not covered in this paper to make the CORE claims of this benchmark justified..

E. DIAGNOSTIC ENGINE

The intention of the Diagnostic Engine was that it should carry out a stratified analysis of the seven baseline systems on all four stratum axes outlined in Section III-A, rather than just coming up with an overall score for each of the systems, such as the discovery of a system with an unusually high error rate with highly conjunctive text. The stratified testing of the whole corpus on all four stratum axes will be left for future work, since this is currently undergoing scaling and tagging (Section VI); Section V shows the aggregate leaderboard results for now.

F. PUBLIC LEADERBOARD AND SUBMISSION INTERFACE

In order to extend the implementation from this paper, the metric engine that was described in section III-D is wrapped inside a local leaderboard application, in which the user will submit their own OCR engine and run a test on their submission in the same manner as the tests performed on the seven engines, in one of the two following methods. The API-based method requires a HTTP endpoint URL, the request template and the path into the JSON response that includes the predicted text provided by the submitter. The other model-file based method requires a submitter supplied python file that contains the recognize(image_path) function; the application will import and use this function once per image.

IV. IMPLEMENTATION

The system described in Section III has been instantiated in the shape of a real-world pipeline: a tagged corpus, seven OCR engines constantly evaluated over this corpus, a grapheme-cluster based correct scoring system, and finally a submission system based on the scoring system.

A. DATA INGESTION & STRATIFIED TAGGING MODULE

The current corpus consists of a fixed-seed sample (seed = 42 for consistent purposes) derived from IndicSTR12 [4] combined with a custom-built corpus of photographs of real-life Tamil pages with different lighting conditions, angles of camera shots, papers of different levels of degradation, which were hand-transcribed and validated by another transcriber. Each page was tagged in accordance with the fixed four-axis system before the scoring: matra_present is scored according to whether there are pre-base dependent vowels in the reference

transcription; conjunct_density is scored according to whether virama (pulli) is present in the reference; degradation_type is scored automatically by a Laplacian variance blur detection algorithm and verified manually; and capture_source is provided by the sampling method.

B. MULTI-GENERATION BASELINE EXECUTION MODULE

The seven baselines mentioned in Section III-C are evaluated using their exact setup as given verbatim for reproducibility purposes. A few issues have been encountered and resolved with respect to various engines instead of applying an arbitrary hack. The Gemma 4 26B showed very suspicious error rates in the first trial run because of some issue with hallucination / over-completion that can be fixed through adjusting the anti-hallucination prompt; however, applying the fix to the Gemini 3.5 Flash-Lite led to lower quality results, and hence, it is executed using the original version of the prompt. Tesseract seemed to run in a non-page segmentation mode with no options; thus, the --psm 7 option had to be explicitly set for running it in a page segmentation mode. Loading of the Tamil language model for EasyOCR failed due to mismatch in the number of classes in the checkpoint and configuration of the package; the suitable checkpoint and configuration were chosen to resurrect it.

C. GRAPHEME-AWARE METRIC ENGINE

They are determined based on Unicode extended grapheme cluster (split by \X in regular expression syntax provided by the regex module, which is identical to UAX #29), rather than code points alone, according to Eqs. (1)–(2). Subsequently, the F1 reorder/misread decomposition (Eq. (3)), for its part, reverses the process to reconsider each edit found through the grapheme alignment procedure, marking it as a reorder R if it matches any of the pre-base matra/consonant switches listed in a lookup table; otherwise, a misread M. See Fig. 2.

Reference graphemes: அ · க் · ஷ · ர · ம்

Hypothesis graphemes: அ · க் · ச · ர · ம்

$F1_{\text{misread}} = M/N = 20.0\%, \quad F1_{\text{reorder}} = R/N = 0\%$

Fig. 2. Grapheme-aware F1 reorder/misread decomposition, traced on the அக்ஷரம் → அக்சரம் example: one misread edit (ஷ→ச), no reorder edits.

D. DIAGNOSTIC ENGINE

At this stage, after executing the process, CER, WER, and the F1 decomposition are subsequently added up individually for each engine for each stratum axis, according to IV-A, provided a sufficient number of samples per stratum is surpassed; those that do not achieve that will be intentionally discarded and combined into an adjacent stratum. This strategy was conceived and implemented (though not fully as described in III-E) along all four axes of the corpus.

E. RELEASE MODULE

Both the corpus and the scoring harness are being released with conditions imposed by each of the datasets separately, i.e., the components that have an explicit permissive license have been released, while components that do not have any license information have been added for evaluation purposes only until permission to distribute is obtained in writing from the maintainers of the component.

F. LEADERBOARD & SUBMISSION INTERFACE

The leaderboard for the public benchmarking suite mentioned in Section III-F is a straightforward local web application that includes the scoring backend using the same metric engine as in Section IV-C, and a browser interface to display the leaderboard, the accuracy plot for the engine, and a visualization of the predictions made by the engine with the thumbnails along with the ground truth. The new engines are directly placed on the leaderboard, together with seven baseline engines, after submitting the run according to the median CER/catastrophic rate of Table II.

V. RESULTS

A. PILOT VALIDATION ON INDICSTR12 (N = 100)

Grapheme-CER values for all seven baseline systems as described in Section III-C are reported in Table I and Fig. 3, when applied to a fixed seed (seed = 42) set of n = 100 images from IndicSTR12 [4] pilot dataset as outlined in IV-A. The value of CER is calculated using Eq. (1) and the per-image results are averaged over 100 images.

Rank	Engine	Tier	CER (%)	× Best
1	Gemini 3.5 Flash-Lite	Frontier	48	1.0×
2	EasyOCR	Classical	58	1.2×
3	Tesseract (--psm 7)	Classical	80	1.7×
4	Chandra	Frontier	99	2.1×
5	Gemma 4 26B	Frontier	111	2.3×
6	PaddleOCR-VL	Open-VLM	123	2.6×
7	Surya	Open-VLM	498	10.4×

TABLE I — Grapheme-CER by OCR Engine, IndicSTR12 Pilot Sample (n = 100)

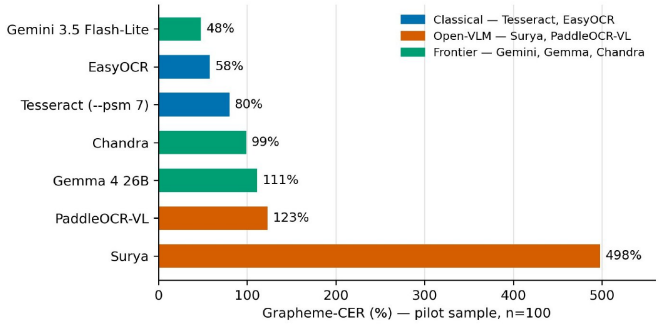


Fig. 3. Grapheme-level CER by OCR engine, IndicSTR12 [4] pilot sample (n = 100), colored by generation tier.

There were two observations from Table I that led to the creation of the stratified diagnostic model in Section III-E. Firstly, the accuracy of the models cannot be predicted by their generation tiers; Gemini (Frontier) is at 48%, while the other two engines within the same tier have much greater accuracies, Chandra at 99% and Gemma at 111%, for an average of 86%

within the tier, which is higher than both Classical engines individually. Secondly, there is considerable variation among the seven engines as the construction of Table I is designed to show.

B. FULL-CORPUS BENCHMARK (N = 644)

After the pilot test, the whole seven-engines benchmark test was run for five Tamil datasets (both natural and artificial samples: hw_words_tamil_sample, indicstr12_sample, synthetic_indicstr_tamil_sample, self_collected_handwritten, and synthetic_r_tamil_sample), for 644 images per engine (except for PaddleOCR-VL – 643 images). The findings are provided in Table II and Fig. 4.

There are two important methodological refinements from the pilot test described in this part. The first one is that CER/WER is counted as a per-image median, and not the mean of Table I. The methodological refinement became necessary in the full corpus test, when two individual images of Gemma 4 26B suffered from a problem of runaway generation (with 38,343 and 32,758 characters being added to the reference transcriptions that contained 6-11 characters each). CER for such images becomes inflated one order of magnitude only by themselves through calculation via mean. CER per-image ≥ 5.0 is considered catastrophic and is reported separately in another column – this is an intended methodological refinement from mean/micro-average options discussed before in this project,The second method improvement is that the results presented now have five datasets instead of one IndicSTR12 pilot sample, effectively increasing each engine's sample size sixfold.

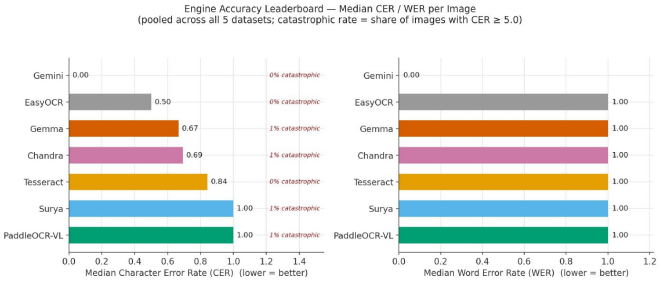


Fig. 4. Full-corpus engine leaderboard: median CER and median WER per image, pooled across all five datasets (n = 644), with catastrophic-rate annotations.

Rank	Engine	Tier	Med. CER	Med. WER	Cat. %
1	Gemini 3.5 Flash-Lite	Frontier	0.00	0.00	0.3%
2	EasyOCR	Classical	0.50	1.00	0.3%
3	Gemma 4 26B	Frontier	0.67	1.00	1.1%
4	Chandra	Frontier	0.69	1.00	1.1%
5	Tesseract (--psm 7)	Classical	0.84	1.00	0.2%
6	Surya	Open-VLM	1.00	1.00	0.6%
7	PaddleOCR-VL	Open-VLM	1.00	1.00	0.8%

TABLE II — Full-Corpus Leaderboard: Median CER/WER and Catastrophic Rate (n = 644, pooled across 5 datasets)

Gemini remains the engine with the most accurate OCR (CER of median = 0.00, which means that most of the predictions made by this engine are absolutely correct), followed by EasyOCR (0.50), Gemma (0.67), Chandra (0.69), and Tesseract (0.84), and the tied engines are Surya and PaddleOCR-VL with the highest median CER (1.00). However, catastrophic error rates remain lower than 1.2% for all the engines, which means that catastrophic errors are not observed in any image but are characteristic for the full set rather than some particular engine. What is worth noting, however, is that the relative ordering within the classical tier varies depending on the scale. Thus, while at the pilot scale Tesseract outperforms EasyOCR (Table I), this is not the case at the full-corpus scale (Table II). Again, the results confirm that the relative performance of a set of seven OCR engines tested on n = 100 cannot be guaranteed to repeat on n = 644. Speaking about WER vs. CER, the pilot data set is relevant because in this case WER is almost binary, since most of the images are words. A related paper [16], which uses the exact same seven OCR engines and the exact same full-corpus test, raises not the question of how many times each engine was wrong but rather the question of what type of error dominated each.

VI. FUTURE WORK

- The complete four-dimensional stratification of diagnosis according to the presence of matra, conjunct density, degradation type, and source of capture (as described in Section III-E) still needs to be performed on the entire corpus because compilation and ground truthing of the corpus is still in process, and that would be the primary output of the next phase of the project.
- A related paper [16] gives the implementation of the F1 reorder/misread decomposition algorithm described in Section III-D, using a five category taxonomy of confusion pairs at full-corpus level, and that will be the platform for further error correlation studies and ASIS/DAEP analysis after their initial pilot phase
- The public leaderboard displayed in Sections III-F and IV-F has been implemented as a local single-user application, and the development of resilience for an online version (submission validation, API mode throttling, automated leaderboard gaming detection) is necessary before this tool can serve as a public benchmarking tool.
- Since the existing literature tends to classify Tamil in the company of its structural counterpart, the Sinhala abugida [8]–[10], it would not be surprising if the stratification scheme and grapheme-oriented metric engine described in this paper could be adapted to other Brahmic scripts as well.

VII. CONCLUSION

The benchmarking framework proposed in this paper for Tamil OCR is: a stratified corpus of real-world documents annotated along four dimensions and a metric engine taking into account grapheme clusters, as well as formulas given in Eqs. (1)–(3), that can distinguish between trivial script serialization errors and genuine errors. Apart from system design, in this paper, we presented preliminary results of IndicSTR12 experiment (n = 100) and a complete benchmarking experiment of seven OCR systems on 644 images taken from five Tamil corpora (both

real-world and artificial), and an interactive leaderboard system was implemented, allowing a new OCR system to be benchmarked with the help of the very same methodology. Stratification results on all the four-axis corpus will be published in later publications. A complementary paper [16] makes use of the results of the benchmark experiment to answer the diagnostic question posed in this paper but left open – why not how much – with the help of confusion pair taxonomy and, as an ongoing work, cross-system error correlations.

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